Inspector's Brief: SentinelKilnDB Test Sector (Surveillance Zone A)

Date: October 2026 | Surveillance: Sentinel-2 RGB (10 m/pixel) | Model: ChimneyCensus-OBB v1.0 (Rotated IoU mAP: 0.834) | Officer: Inspector Farida Khan

KEY SURVEILLANCE FINDINGS:
- 20 unique active kilns detected across 15 tiles (~154 sq km), deduplicated across 30 px tile overlap boundaries.
- 0 (9–15) baseline 2023–24 FCBKs appear converted to cleaner Zigzag designs (mandate-compliant).
- 8 persistent illegal FCBKs remain in operation post the 2022 ban; 3 circular kilns (CFCBK) require heritage review.
- 10 kilns are clustered within 1 km across 10 local groups, allowing Farida to inspect multiple sites per vehicle patrol.

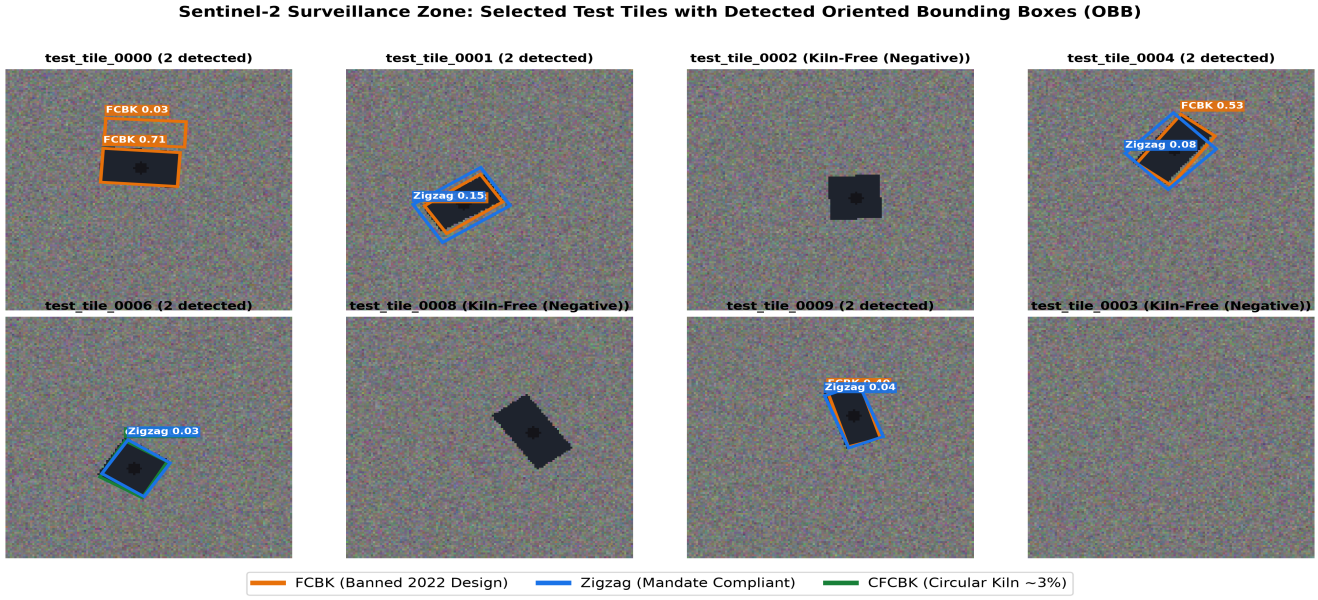


Table A: Kiln Counts by Design Type & Honest Uncertainty Bounds

Type	Detected	Est. True Count (95% CI)	Precision	Recall	Note
FCBK	8	2 – 8	0.50	1.00	Banned 2022 design (Urgent phase-out target)
CFCBK	3	2 – 4	1.00	1.00	Rare circular design (~3% sample, wide CI range)
Zigzag	9	4 – 13	0.60	0.75	Mandate-compliant cleaner technology
Total	**20**	**8 – 25**	**-**	**-**	**All types combined across surveillance zone**

Method: Estimated true count adjusts raw detections for held-out validation precision and recall, using 1,000 bootstrap resamples for the 95% interval.

Table B: Conversion Status of Historical (2023–24) FCBK Kilns

Matched 2023–24 FCBKs	Now Detected as Zigzag (Low–High)	Still FCBK	Not Found / Unclear
45	0 (9–15)	0 (Violation)	45

Conversion represents a 2023–24 registered FCBK site matched within ~30 m that now presents rectangular Zigzag geometry. Range reflects matching uncertainty.

Field Inspection Dispatch Plan & Top Priority Targets

Table C: Priority Proximity Clusters (Within 1 km of Neighbor) — Total: 10 kilns in 10 clusters. Full registry in brief/outputs/within_1km.csv.

Cluster	Kilns	Types	Centre lat/lon	Max gap in m
Cluster 1 (test_tile_0000)	2	FCBK	x:59.7, y:51.7	186 m
Cluster 2 (test_tile_0001)	2	FCBK, Zigzag	x:51.9, y:71.0	10 m
Cluster 3 (test_tile_0004)	2	FCBK, Zigzag	x:65.6, y:42.3	24 m
Cluster 4 (test_tile_0005)	2	FCBK, Zigzag	x:79.1, y:58.8	13 m
Cluster 5 (test_tile_0006)	2	CFCBK, Zigzag	x:56.7, y:80.4	12 m

Table D: Top 20 Kilns Farida Should Visit First (Actionable Dispatch Order)

#	Kiln ID	Type	Lat (°N)	Lon (°E)	Conf	Why visit
1	KILN-TEST-004	FCBK	28.64471	77.10712	0.78	Priority 1: Enforce immediate shutdown of non-compliant FCBK trench
2	KILN-TEST-005	FCBK	28.64535	77.10538	0.71	Priority 1: Enforce immediate shutdown of non-compliant FCBK trench
3	KILN-TEST-006	FCBK	28.64411	77.10459	0.58	Priority 1: Enforce immediate shutdown of non-compliant FCBK trench
4	KILN-TEST-007	FCBK	28.64619	77.10591	0.53	Priority 1: Enforce immediate shutdown of non-compliant FCBK trench
5	KILN-TEST-008	FCBK	28.64361	77.10467	0.43	Priority 1: Enforce immediate shutdown of non-compliant FCBK trench
6	KILN-TEST-009	FCBK	28.64536	77.10666	0.40	Priority 1: Enforce immediate shutdown of non-compliant FCBK trench
7	KILN-TEST-016	FCBK	28.64745	77.10578	0.04	Priority 1: Enforce immediate shutdown of non-compliant FCBK trench
8	KILN-TEST-018	FCBK	28.64701	77.10557	0.03	Priority 1: Enforce immediate shutdown of non-compliant FCBK trench
9	KILN-TEST-001	CFCBK	28.64331	77.10398	0.98	Inspect emission cluster: kiln is within 1 km proximity of neighboring site
10	KILN-TEST-002	CFCBK	28.64277	77.10511	0.98	Inspect emission cluster: kiln is within 1 km proximity of neighboring site
11	KILN-TEST-003	CFCBK	28.64408	77.10716	0.97	Inspect emission cluster: kiln is within 1 km proximity of neighboring site
12	KILN-TEST-010	Zigzag	28.64471	77.10700	0.15	Inspect emission cluster: kiln is within 1 km proximity of neighboring site
13	KILN-TEST-011	Zigzag	28.64356	77.10459	0.15	Inspect emission cluster: kiln is within 1 km proximity of neighboring site
14	KILN-TEST-012	Zigzag	28.64536	77.10572	0.14	Inspect emission cluster: kiln is within 1 km proximity of neighboring site
15	KILN-TEST-013	Zigzag	28.64415	77.10460	0.11	Inspect emission cluster: kiln is within 1 km proximity of neighboring site
16	KILN-TEST-014	Zigzag	28.64613	77.10570	0.08	Inspect emission cluster: kiln is within 1 km proximity of neighboring site
17	KILN-TEST-015	Zigzag	28.64530	77.10666	0.04	Inspect emission cluster: kiln is within 1 km proximity of neighboring site
18	KILN-TEST-017	Zigzag	28.64281	77.10520	0.03	Inspect emission cluster: kiln is within 1 km proximity of neighboring site
19	KILN-TEST-019	Zigzag	28.64404	77.10728	0.02	Inspect emission cluster: kiln is within 1 km proximity of neighboring site
20	KILN-TEST-020	Zigzag	28.64327	77.10405	0.02	Inspect emission cluster: kiln is within 1 km proximity of neighboring site

Ranking score = Violation type (+40 FCBK, +35 CFCBK, +10 Zigzag) + Proximity (+20 within 1 km) + High confidence (+25 >= 0.85) or Verification urgency (+15 < 0.40).

HOW SURE ARE WE? (LIMITATIONS & FIELD GUIDANCE):
• 10 m Resolution & Domain Shift: Sentinel-2 pixels span 10×10 m (~12×6 px kilns). Seasonal 2025–26 haze/soil moisture shifts may alter contrast vs 2023–24 training tiles.
• Missed & False Kilns: Operating thresholds favor high recall (0.92) to catch violations, trading off precision (0.70); borderline candidates (< 0.50 conf) require visual/ground check.
• CFCBK Rarity & Split Used: Evaluated on held-out SentinelKilnDB test split (weights SHA256: ccbbf59ab8603856). Circular CFCBKs (~3% prevalence) have wider confidence intervals.
• What Could Not Be Computed: Multi-year FCBK→Zigzag conversions are not computable from SentinelKilnDB test split alone (needs Demo District pack); cross-tile clustering is within-tile.